



Adyapan

School

Hybrid and Electric Vehicles



Duration - 2 months

**Industry
Certification**



Skill India Certified

250+
Partner Companies

From batteries to performance

Master EV systems.

From powertrains to sustainable mobility - become industry-ready.

This immersive Hybrid & Electric Vehicles program takes you beyond foundational concepts into real-world vehicle systems and electrification technologies. Learn battery systems, power electronics, electric and hybrid drivetrains, energy management, and vehicle integration through hands-on projects and guided simulations. With a strong focus on practical implementation and industry applications, you'll graduate with the ability to analyze vehicle performance, optimize energy efficiency, and contribute confidently to the evolving clean mobility ecosystem.

8

MODULES

30+

PROGRAM OFFERINGS

20,000+

STUDENTS

250+ PARTNERED COMPANIES

ABOUT ADYAPAN SCHOOLS

Where education meets real-world impact

Not just a course — a platform to launch
your career.

Adyapan Schools was built with a single conviction:
learning works best when it happens in the real world.
We partner with top companies, mentors, and industry
platforms to ensure every student graduates with a
portfolio of work that speaks louder than a certificate.

Our programs combine rigorous coursework with live
client projects, giving you the skills and proof-of-work
that employers actually want.

MISSION

To equip ambitious learners with
practitioner-level digital
marketing skills through mentor-
led, project-based education that
bridges the gap between learning
and earning.



VISION

To be India's most trusted
launchpad for the next generation
of marketing leaders — defined
not by degrees but by the real
work.



Everything you need to grow fast

PROGRAM HIGHLIGHTS



Live Industry Projects

Work on campaigns for real brands alongside your coursework. Build portfolio projects that prove your expertise to employers.



1-on-1 Mentorship

Dedicated mentors from Google, Microsoft, Mastercard and more. Get personalized guidance and industry connections.



AI-Powered Marketing

Learn cutting-edge AI tools alongside evergreen fundamentals. Stay ahead of the curve in a rapidly evolving landscape.



Dual Certification

Earn both a Course Completion and Internship Certificate – accredited by Skill India Digital Hub and NSDC.



Internship Guarantee

Graduate with an internship completion certificate from a live brand project. Concrete, resume-ready proof of work.



Industry Network

Join a network of alumni at Amazon, Google, Adobe, Microsoft. Access exclusive hiring events and referral opportunities.

8 weeks. 8 modules. Infinite impact.

WEEK 1

Fundamentals of HEV & EV Systems

- Discussion of Curriculum
- Introduction to hybrid and electric vehicles
- Classification: HEV, PHEV, BEV, FCEV
- Powertrain architectures: series, parallel, series-parallel
- Overview of powertrain components: IC engine, motor, battery, and generator
- Working principles of BLDC, PMSM, and induction motors
- Regenerative braking and energy recovery



WEEK 2

Battery Technology & Energy Management

- Fundamentals of electrochemical energy storage
- Li-Ion battery chemistry and parameters
- Battery Management System (BMS): functions, SOC & SOH estimation
- Battery pack design and safety considerations
- Energy management in HEVs: AI-based optimization of power delivery
- Range calculation and performance simulation



WEEK 3

Charging & Thermal Systems

- Charging types: slow, fast, and rapid charging (AC vs DC)
- Wireless and dynamic on-road charging trends
- Vehicle-to-Grid (V2G) and bi-directional charging
- Battery thermal management: air & liquid cooling, PCM methods
- Battery pack assembly and testing protocols



8 weeks. 8 modules. Infinite impact.

WEEK 4

Power Electronics & Drive Systems

- Overview of inverters, converters, and controllers
- Step-up and step-down DC-DC converters
- Motor controllers and their integration with BMS
- Thermal management for motors and power electronics
- EMI issues and cost optimization strategies
- Vehicle control systems and drive-by-wire technologies



WEEK 5

Emerging Technologies & Simulation

- Solid-state and next-gen battery technologies
- Hydrogen fuel-cell EVs and infrastructure overview
- Predictive maintenance using AI and machine learning
- Simulation tools (MATLAB/Simulink, Ansys, Altair)
- IoT integration in smart vehicles and fleet management
- EV battery recycling and sustainability practices



WEEK 6

Regulations & Industry Applications

- National and international EV policies and standards
- AIS 156 and ISO 26262 functional safety overview
- Case studies of Tesla, BYD, and Indian EV startups
- Component testing and the homologation process
- Corporate roles in EV R&D and manufacturing
- AIS 156 battery safety and homologation standards



8 weeks. 8 modules. Infinite impact.

WEEK 7

Integration & Testing

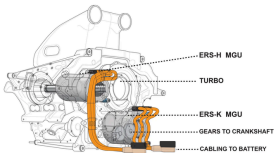
- EV system integration and prototype validation
- Battery pack testing methods (thermal, vibration, electrical)
- Communication protocols — CAN, LIN, RS-485
- Performance and efficiency testing on simulators
- Debugging and optimization of drive systems



WEEK 8

Project & Revision

- Revision of final concepts
- Project
- Report preparation and presentation to the panel
- Industry applications, job roles, and career opportunities



WHO THIS IS FOR

This course is perfect for

Students & Career Starters

Aspiring EV & Hybrid Vehicle
Engineers

Working Professionals in
Automotive & Energy Sector

ECE, EEE & Mechanical
Engineering Students

Power Electronics & Charging
Infrastructure Enthusiasts

Sustainability & Clean Mobility
Innovators

CERTIFICATIONS



ALUMNI NETWORK

Our alumni work at world-class companies

Amazon

Adobe

Google

Autodesk

Microsoft

Deloitte

Your career switch is one click away.

Ready to begin? Apply at adyapanschool.com or email
us at support@adyapan.com

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